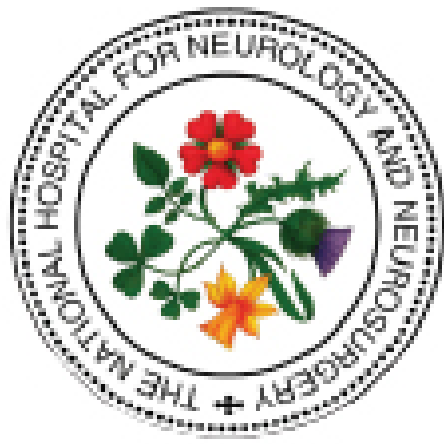




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU) EARLY REHABILITATION in NEUROSCIENCE PATIENTS

Any potentially conflicting issues (e.g. ventilation wean vs rehabilitation) must be discussed by MDT and rationale for any decisions documented and communicated

STAGE 1 NEW PATIENT INITIAL ASSESSMENT

NCC TEAM HANDOVER

REVIEW

Medical Records

Admission diagnosis
Previous medical history

CONSIDER

Individual patient restrictions

Check **SPINAL & ANGIO**
post-procedure instructions

Planned surgery / investigations

CONSIDER

A B C D E

Respiratory status
Cardiovascular status
Neurological status

COMPLETE

Initial Physiotherapy
Treatment Proforma

Does patient
meet criteria
for early
rehabilitation?

NO

Defer mobilisation or
discuss with senior
physiotherapist or
NCC team

Document plan and
continue to review

UNSURE

Discuss with
NCC team and
senior
physiotherapist

Early rehabilitation and mobilisation criteria

Is there sufficient cardiovascular reserve?

- Resting HR <50% age predicted maximal HR
 - HR >40 or <130bpm
 - BP <20% variability recently
 - MAP >65 or <110mmHg
- ECG normal (no evidence of MI / arrhythmia)
- Other major cardiac conditions excluded

Is there sufficient respiratory reserve?

- FiO₂ <.8, spO₂ >90% and <4% recent decrease in spO₂
- Respiratory pattern satisfactory
- 1. Mechanical ventilation able to be maintained during treatment

Are all other factors favourable?

- Weight able to be managed
- Safe environment
- Appropriate staffing and expertise
- Appropriate equipment
- Patient consent - where appropriate

MDT agree
to mobilisation
and early
rehabilitation

YES

Proceed to plan
rehabilitation
session →
STAGE 2

NO

Acute subarachnoid haemorrhage (SAH) patients should be on Bed rest until cause investigated +/- treated
If spinal patient check stability of spine

A and B

What mode of ventilation and how much oxygen does patient require?

Evidence of aspiration?

C

Is patient requiring high level of cardiovascular support (noradrenaline / labetalol)?

D

Is ICP / CPP stable?
New neurological deficit?
Recent seizure?

E

Infection
Intravenous / arterial lines
External ventricular drain
Chest drain / Wound drain
Diarrhoea
Nausea / vomiting
NG feed - high aspirates
Pressure ulcer

STAGE 2 EARLY REHABILITATION TREATMENT PLAN / GOALS

Patient stable during
rehabilitation session?

NO

STOP
TREATMENT
AND MAKE
NEW PLAN

UNSURE

DISCUSS
WITH SENIOR
PHYSIOTHERAPIST
OR NCC TEAM

YES

CONTINUE
TREATMENT
AND MAKE
STEPWISE
PLAN

STEPWISE APPROACH to MOBILISATION

- Facilitated rolling
- Facilitated to edge of bed
- Hoist to chair

Ensure **TIME LIMIT** prescribed

Sacral pressure ulcer or bowel management system (BMS) do not preclude sitting out of bed per se – but requires individual patient assessment – limit seating to <2h with regular review of pressure areas after every session

Tilt table

Standing frame

Transfers

Sit to stand / stand to sit facilitation

Gait

Thera bike / CV training

- Continue 24 hour positioning implementation
- Tone management
- Splinting programme
- Additional outcome measures
- Referral to OT as required

Rehabilitation treatment plans
and goals agreed by MDT

These should be documented
and reviewed daily / as required
on MDT ward round

Additional considerations prior to and during rehabilitation and mobilization

ENVIRONMENT & PATIENT SPECIFIC CONSIDERATIONS

- INFUSION LINES / IV DRUGS:**
Can anything be disconnected?
- EXTERNAL VENTRICULAR DRAIN:** Nurse must monitor drain and advise on how long it can be clamped?
- ET TUBE / TRACHEOSTOMY:**
Tubes **must be physically supported** at lips / neck to prevent dislodging
- VASOSPASM** in treated aneurysmal SAH: D/w MDT

Noradrenaline / Labetalol not necessarily a contraindication to mobilising provided patient stable and not requiring high levels (>0.2mcg/kg/min Noradrenaline and >9mg/h Metaraminol) and must be discussed with MDT and nurse present to monitor if plan to mobilise

Does patient require analgesia prior to treatment?

Does sedation need to be altered prior to treatment?

If spinal patient, do they require an abdominal binder / brace / collar?

Does patient require ephedrine to support BP?

Consider patient's level of co-operation / cognition / delirium

Decision pathways
Additional considerations